Harsh Bordekar

AI Scientist – Physics Models

Physics-Informed ML for Simulation, Surrogate Modelling & Uncertainty Quantification

PhD Candidate, DLR & TU Braunschweig — Aerospace Composites & Structural Simulation

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Profile

Simulation and machine-learning research engineer with 7+ years developing physics-based models for aerospace structures, including 3+ years applying ML and surrogate modelling directly to simulation problems. PhD work builds a probabilistic, physics-based multiscale surrogate framework for CFRP composites – coupling microstructure-informed mesoscale damage models with stochastic spatial defect distributions to propagate failure from ply to structural scale, replacing computationally expensive full-scale FEM with fast, physics-consistent predictions – alongside a certification-aligned failure criterion that the framework is evaluated against. Runs large-scale simulation campaigns on HPC/SLURM using domain-specific solvers (ABAQUS, Nastran/Patran) to generate the high-fidelity data these models depend on, and applies Monte Carlo uncertainty quantification to characterise prediction confidence ahead of design decisions – direct, hands-on experience with the UQ/out-of-distribution estimation this role treats as a first-class concern. Also built a physics-based CNN surrogate for on-demand computational homogenization, exposed through a tool-calling interface, replacing full-scale FEM homogenization runs with fast inference. Comfortable across the applied-ML stack (PyTorch, TensorFlow, scikit-learn) and in Python/Linux/HPC environments day to day, with a track record of verifying and technically reviewing externally produced simulation models against defined acceptance criteria, and of explaining simulation and ML concepts to both engineering and non-technical audiences.

Core Competencies: ML for Simulation & Surrogate Modelling, Physics-Informed Machine Learning, Multiscale / Multi-Fidelity Damage Modelling, Uncertainty Quantification & Out-of-Distribution Estimation, High-Fidelity Simulation Data Generation, FE Solvers (ABAQUS, Nastran/Patran), HPC-Scale Simulation Campaigns (SLURM), Model Verification & Validation, Physics-Based CNN / Surrogate Modelling, Explainable AI, Python

Professional Experience

Structural Analysis / Simulation Engineer — DLR & TU Braunschweig

Physics-informed ML for simulation & surrogate modelling (PhD) — *Feb 2021 - Present*

- (PhD) Developed a probabilistic, physics-based multiscale surrogate framework for leakage-onset prediction in CFRP hydrogen storage tanks, coupling microstructure-informed mesoscale damage models with stochastic spatial defect distributions to propagate failure across ply, laminate, and structural scales – replacing **[x]** full-scale FEM runs with a physics-consistent surrogate and estimating probability of failure directly from the simulation-generated dataset.
  Tools: Python, ABAQUS, FEM, HPC/SLURM, Multiscale Modelling
- (PhD) Derived a certification-aligned leakage failure criterion mapping combined strain/load states to leakage onset, bridging progressive-damage models and system-level tank simulations and providing the physics-based evaluation standard against which the surrogate framework's coverage, accuracy, and failure modes are assessed.
  Tools: ABAQUS, Python, FEM post-processing using C++
- Performed Monte Carlo uncertainty quantification over the multiscale simulation outputs to generate leakage-probability-vs-strain relationships, characterising prediction confidence and enabling risk-informed design decisions for next-generation CFRP pressure vessels – direct UQ/OOD experience applied to physics-simulation outputs.
  Tools: Python (SciPy, NumPy), Monte Carlo methods, ABAQUS
- Ran large-scale thermal and structural simulation campaigns on HPC infrastructure to evaluate thermal-protection effectiveness for the Callisto reusable-rocket programme's metal upper housing bay, using domain-specific solvers to generate high-fidelity simulation data under production-scale compute constraints.
  Tools: ABAQUS, SLURM (HPC), Python, Nastran, Patran
- Built a physics-based CNN surrogate for microstructure-informed computational homogenization, wrapped in an agentic, tool-calling interface that accepts natural-language queries and microstructure images, retrieves context via a hierarchical RAG pipeline over **50** materials documents, and returns homogenized/effective material properties – replacing full-scale FEM homogenization runs (hours) with on-demand inference (seconds). Retrieval/prediction faithfulness validated at **93**% against ground-truth homogenization results on a held-out set of more than **100** cases.
  Tools: Python, Physics-Based CNN, Hierarchical RAG, LLM Agent Framework, Image Processing
- Verified and technically reviewed externally and student-produced simulation models, meshes, and results against defined methods and acceptance criteria, diagnosing failure modes arising from data gaps or modelling limitations and issuing documented sign-off-style findings.
  Tools: ABAQUS, Nastran, Python

- Instructed 60+ Master's students in FEM theory and ABAQUS-based CFRP modelling, covering Continuum Damage Mechanics (CDM), Progressive Damage Analysis (PDA), and Cohesive Zone Modelling (CZM) – demonstrated ability to communicate simulation and ML concepts to both technical and non-technical audiences.
  Tools: ABAQUS, Automation using Python

Research Intern, Metallic Fatigue & Damage Tolerance — DLR-Braunschweig
Explainable AI & physics-based fatigue/defect-tolerance assessment — *Nov* 2019 - *Jan* 2021

- Developed a CT-based, explainable-AI defect-detection framework achieving **91**% classification accuracy (**89** precision / **91** recall) on **1200** CT scans of additively manufactured titanium parts, cutting manual review time to <**0.5 sec per image**, with SHAP, Grad-CAM and LIME outputs used to support engineering sign-off. *(Published: Journal of Intelligent Manufacturing, 2025.)*
  Tools: Python (scikit-learn, TensorFlow), SHAP, Grad-CAM, LIME, CT image processing
- Assessed fatigue and defect tolerance of aerospace-grade titanium components using small-defect fatigue models (Murakami, El-Haddad), coupling defect measurements with numerical simulation to translate physics-based fracture-mechanics criteria into allowable-defect acceptance limits.
  Tools: Small-defect fatigue models (Murakami, El-Haddad), Python (NumPy/SciPy), FEM (ABAQUS)

Student Research Assistant — Leibniz University Hannover
Institute of Static and Dynamic — *Sep* 2018 - *Oct* 2019

- Developed an ML framework for automated detection and quantification of fiber undulations in unidirectional CFRP laminates from image data, generating spatially resolved defect metrics used directly as high-fidelity inputs for probabilistic structural simulation models assessing manufacturing-induced variability.
  Tools: MATLAB, Python (scikit-learn, TensorFlow), image processing, NumPy/SciPy

Skills Summary

ML for Simulation & Surrogate Modelling	Physics-Informed Machine Learning | Multiscale / Multi-Fidelity Damage Modelling | Physics-Based CNN Surrogates | Computational Homogenization
Uncertainty & Evaluation	Uncertainty Quantification (Monte Carlo) | Verification & Validation | Physics-Based Evaluation Against Certification Criteria | Failure-Mode Diagnosis
Simulation Solvers & HPC	ABAQUS | Nastran/Patran | Linear & Non-linear FEM | HPC / SLURM (Large-Scale Simulation Campaigns) | Analytical Stress Methods (Niu, Bruhn)
ML & AI	PyTorch | TensorFlow | Scikit-learn | Explainable AI (SHAP, Grad-CAM, LIME) | Hugging Face Transformers (Fine-tuning)
LLMs & Agentic Systems	Retrieval-Augmented Generation (RAG) | LangChain | LangGraph | Model Context Protocol (MCP) | Tool-Calling
Programming Languages	Python | C++ | FORTRAN | MATLAB
Data Analysis & Scientific Computing	NumPy | Pandas | Matplotlib | Seaborn | Plotly | SALib | SciPy
APIs & Backend Development	RESTful APIs | FastAPI | JSON
Version Control & Reproducibility	Git | Docker | Linux
Application Development	VS Code | JupyterLab | Google Colab | Streamlit | Gradio
Languages	English (C1) | German (B1)

Awards and Education

2nd Prize: Best Paper Award — DLR (German Aerospace Center)
2026

3rd Prize: Best Paper Award — DLR (German Aerospace Center)
2024

PhD Candidate, Computational Methods in Engineering — DLR & TU Braunschweig, Germany
Feb 2021 – 2026 (expected)

Dissertation: Physics-Informed, Multiscale Machine Learning for CFRP Composite Structures

M.Sc. Computational Methods in Engineering — Leibniz University Hannover, Germany
April 2018 – Dec 2020

Focus: Numerical Methods, Scientific Machine Learning, Deep Learning

Selected Publications

1. **Bordekar, H.**, et al. "Microstructure-informed Probabilistic Multiscale Modelling of CFRP via Machine-Learning-Derived Stochastic Volume Elements." *Manuscript in preparation*, 2026.
2. **Bordekar, H.**, et al. "MicroStructAgent: A Physics-Informed Agentic AI Framework for Computational Homogenization and Quality Assurance of Fiber Composite Laminates." *Manuscript in preparation*, 2026.
3. **Bordekar, H.**, et al. "Microstructure Characterization of Fiber Composite Laminate Using eXplainable Artificial Intelligence." *Journal of Composite Materials*, vol. 60, no. 1, 2026, pp. 3–25. DOI
4. **Bordekar, H.**, et al. "eXplainable Artificial Intelligence for Automatic Defect Detection in Additively Manufactured Parts Using CT Scan Analysis." *Journal of Intelligent Manufacturing*, vol. 36, no. 2, 2025, pp. 957–974. DOI

References

Prof. Dr. Christian Huehne	Abteilungsleiter Funktionsleichtbau. Deutsches Zentrum für Luft- und Raumfahrt. Institut für Systemleichtbau
Prof. Dr. Oliver Voelkerink	Head of Institute., IMA, TU-Braunschweig
Dr. Nicola Cersullo	R&T Project Manager @ Airbus Hamburg